

Semantic Secrets: Private Authentication from Reconstructable Visual Semantics

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Abstract

Most authentication requires exact secret reproduction or exact cryptographic possession. This planning manuscript studies a different primitive: approximate semantic reconstruction, in which independently reconstructed prompts and images may yield compatible canonical semantic credentials. The resulting fuzzy predicate creates a complete acceptance region; legitimate variation must be accepted without making realistic or AI-assisted semantic alternatives too easy to reach. Even a discriminative predicate raises a separate privacy problem if its graph, policy, record, or transcripts enable inversion, offline candidate validation, adaptive probing, or cross-domain linkage. We therefore ask whether reconstructable visual semantics can be stable and discriminative, whether the induced region survives finite-budget attacks, and whether the frozen predicate can be privately verified. This is not a completed system or a human-memory study. P8 narrowed the prospective C1–C4 contribution after finding prior art for the component mechanisms. P9-v2 then tested two frozen local extractors that each had to infer a scene and author complete exact credential JSON; each produced an invalid graph on its first formal validation fixture. One invalid among 32 caps best-case schema validity at $31/32 = 96.875\%$, below the preregistered 98% requirement. Gate V2-A failed and full semantic accuracy was not estimated. P9-v3A preregistered a modular observation/compiler architecture, and a prospective v3.1 audit selected EGTR as the graph-native comparator while freezing resource and support rules before output. No v3 weight, capability image, inference, authentication, or cryptographic experiment has run. P10 remains blocked.

1 Problem and scope

Exact-secret authentication can be idealised as accepting a singleton, $\mathcal{A}(s) = \{s\}$. Semantic authentication replaces exact reproduction with approximate reconstruction: different prompts and visual realizations of one intended concept may need to authenticate as equivalent. It therefore induces a fuzzy set $\mathcal{A}(S) = \{S' : \text{Accept}(S, S') = 1\}$. Benign reconstructions must vary, yet every tolerated variation can enlarge the set an attacker may submit. Too strict a predicate rejects independent same-concept reconstructions; too tolerant a predicate gives attackers more accepted alternatives.

This yields a central security question: can independently reconstructed visual concepts produce stable and discriminative canonical credentials whose complete acceptance region remains sufficiently resistant to realistic and AI-assisted guessing? It also yields a separate privacy question. A deterministic graph, policy, protected record, or transcript may reveal semantic content, support database-only candidate testing or adaptive probing, and link reuse across accounts or services. Secure computation may hide the comparison while leaving the accepted region easily enumerable; good discrimination may coexist with template leakage.

We study a prospective client-side pipeline in which an intended concept is reconstructed through free natural language, a local generator produces an independent visual realization, and local perception plus deterministic compilation produce canonical semantics. The strongest target sends neither prompt, image, observation, plaintext graph, nor raw embedding to the authentication infrastructure. The generator serves as a semantic reconstruction and normalisation boundary, not as a memorability device.

No participant study is planned. Researcher-authored concepts and reconstructions can measure only technical semantic reconstruction stability. They cannot establish memorability, recall, usability, preference, accessibility, natural credential choice, real-world authentication time, superiority to passwords or passkeys, or that images are easier to remember. The primitive is not presumed to replace passwords: standalone, second-factor, policy-constrained, restricted-use, negative/measurement, and stop outcomes remain possible.

2 Reconstruction media are not credentials

Let C denote an intended semantic concept and P a free-form reconstruction interface. Different expressions may aim at the same concept—for example, “a cat beside a red car” and “a feline next to a crimson automobile.” The hypothesis is not exact prompt reproduction. With generator G and randomness r ,

$$I = G(P, r), \quad (1)$$

where I is a transient, independently generated reconstruction medium. Pixels do not contribute credential entropy merely because generation is stochastic. A versioned modular pipeline produces probabilistic visual evidence

$$O = \text{Observe}(I), \quad (2)$$

and a deterministic semantic compiler produces the security-sensitive representation

$$S = C(O) \quad (3)$$

or a typed failure. Models infer bounded evidence; code owns schema, identifiers, duplicates, inverse normalization, derived counts and geometry, ordering, and serialization. Thus the prompt is an interface, the image is a transient medium, observations are evidence, and the canonical graph is the representation used by the predicate. Operationally, the security surface is the complete set of candidate semantics the verifier accepts.

This framing differs from treating direct text embeddings or a single generated image as the authentication object. It also creates a falsifiable condition: if independent generation and fresh extraction do not preserve security-relevant structure, the visual reconstruction hypothesis fails.

3 Canonical policy-aware semantics

We provisionally represent S as a typed graph

$$S = (V, E_S, A, C_S), \quad (4)$$

where V contains entities, E_S typed actions/relations, A attributes, and C_S counts and scene/context facts. A deterministic, versioned policy derivation function produces

$$\Pi(S) = (M, T), \quad (5)$$

where M contains mandatory security anchors and T contains secondary facts with bounded tolerance. The primary non-human study will not ask users to nominate M or T . Selection must be frozen from independent semantic rationale, training-only statistics, extractor reliability, attacker-frequency evidence, and protocol compatibility.

A P8-frozen reference functionality has the form

$$\text{Accept}_{\text{ref}} = \text{MandatoryOK}(M, M') \wedge [\text{TolSim}(T, T') \geq \tau], \quad (6)$$

The v3 observation/compiler contract is specified in docs/formal_specification_v3.md. Its broad candidate language L_{visual} contains entities, attributes, actions/interactions, derived geometry/counts, and scenes. A type enters $L_{\text{cred}} \subseteq L_{\text{visual}}$ only after independent capability evidence passes in both controlled and naturalistic strata. Actions and interactions are independently gated: neither is required nor silently discarded. The M/T notation and policy split are preserved but are not implemented or tuned in P9-v3A and are not claimed as standalone novelty.

4 Acceptance-region security

For an enrolled state, define the accepted region

$$\mathcal{A}(S, M, T) = \{(S', M', T') : \text{Accept}_{\text{ref}} = 1\}. \quad (7)$$

An attacker succeeds by finding any member, not by recovering the enrolled prompt, image, or exact graph. Acceptance-region reachability is therefore the security object rather than a usability feature. We distinguish:

- K_0 : generic public knowledge and random candidates;
- K_1 : population or frequency knowledge without target facts;
- K_2 : partial target-specific semantic information;
- K_3 : strong near-secret information.

For strategy Q , budget B , knowledge K_i , and observable view V , the primary quantity is

$$\text{Succ}(Q, B, K_i, V) = \Pr[\exists j \leq B : q_j \in \mathcal{A}(S, M, T)]. \quad (8)$$

The evaluation should report success@1, @5, @10, and @B, guesses to success, duplicate policy, uncertainty, and cost. Random FAR is only one distributional statistic. Frequency, targeted neighbours, partial knowledge, LLM/VLM/generator assistance, and adaptive semantic search may produce different risk. AI-assisted password guessing is established prior art [23]; this project must contribute a semantic acceptance-region analysis rather than the general idea of using a model to guess.

5 Privacy target

The protected record is abstractly

$$R = \text{Protect}(S, M, T, \text{account}, \text{domain}, \text{version}). \quad (9)$$

A later protocol should reproduce the frozen plaintext decision while revealing no more than its declared output and metadata. Candidate goals are:

- template confidentiality against semantic recovery or accepted-equivalent reconstruction;
- resistance to cheap database-only candidate validation under an exact stolen-record view;
- policy privacy for anchor identities, operators, weights, thresholds, and graph shape;
- unlinkable, domain-separated records and transcripts;
- transcript minimisation, replay/session binding, and explicit adaptive-query leakage;
- containment under named database, server, key/share, collusion, and total-compromise states.

Established fuzzy extractors [4], fuzzy aPAKE [5], OPRFs [3], OPAQUE [2], exact PSI [6, 22], fuzzy PSI/private biometric matching [1, 26], committed PSI [7], private fuzzy-record computation [9], private graph operations [13], and homomorphic vector matching [16] are candidate foundations, not novel primitives. Raw learned representations are not automatically private; embedding leakage is an explicit concern [24]. Protected-template unlinkability and irreversibility must be attacked rather than inferred from an encoding [8].

Semantic correctness and cryptographic privacy are independent. FAR/FRR do not prove template, policy, transcript, or linkage privacy; private computation does not repair an easily guessed acceptance region.

6 Closest work and VSA boundary

VSA [11] is the closest mandatory baseline. It enrolls a reference image, extracts objects, CLIP-derived attributes, quantities, visibility, and 2x2 quadrant location, and lets a user select two or three semantic facts plus quantity operators. Authentication uses an arbitrary satisfying image and an additional password. VSA serialises canonical semantic tokens, evaluates Flexible Range Logic conjunctions, hashes most semantic fields together with the password using SHA-256, and retains quantity/operator metadata separately.

These features remove image-independent visual-semantic authentication, VLM extraction, canonical tokens, object, attribute, count, and quadrant rules, flexible quantity policies, conjunction, and user-selected semantic policy from our novelty space. We require two fair baselines: (i) VSA's semantic-policy component for functional discrimination and (ii) the full semantic-plus-password hash architecture for security comparison. We must not compare a semantic-only system directly with VSA's claimed combined semantic-plus-password space as though the credentials were equivalent.

VSA calls its hash binding server-opaque and explicitly weaker than strict zero knowledge. It acknowledges small semantic inputs and fast SHA-256 offline guessing, recommending Argon2id with a user-specific salt. Its COCO-based positive FRR samples

are automatically selected from cached detections rather than independently re-inferred. It reports no human study and leaves Semantic Relationship Logic to future work. It does not target private/unlinkable verification.

SemanticLock uses semantically linked image tokens and story structure for graphical authentication [20]. PassStyles [18] and Omokage [19] use StyleGAN-generated changing face candidates and recognition criteria, preventing broad “first semantic”, “first generative”, or “first changing-image” authentication language. Adaptive score-guided verifier search is likewise established [17]. The P8 review found no verified source combining free-language independent text-to-image reconstruction and fresh typed extraction with a system-derived policy, finite-budget semantic attacks, and construction-specific private/domain-separated verification. Gate V2-N passed for that scoped question only; this is not an exhaustive absence or first claim.

The v3 shortlist is grounded in prior visual-understanding work. Grounding DINO produces open-set, language-conditioned boxes [15]; SigLIP 2 provides contrastive vision-language encoders for closed-label scoring and localization-oriented transfer [25]; and EGTR exposes object, relation, and connectivity predictions from a DETR-based graph model [10]. SGTR [14] remains the historical v3.0 graph candidate. The newer autoregressive ROBIN/Synthetic Visual Genome path [21] was audited but rejected because generated graph text and semantic label mapping would reintroduce structured-generation risk. Visual Genome illustrates the dense annotation problem and its dataset limitations [12]. These are established perception components, not authentication or privacy contributions.

7 Research questions

Together these questions ask whether technical reconstruction works, whether its necessary tolerance creates an acceptable attack surface, whether that predicate can be evaluated privately, and whether the trusted-client design is computationally practical. They do not evaluate human memory or claim password replacement.

RQ1—Reconstruction. Can independently generated images representing the same intended visual concept produce stable canonical semantic credentials?

RQ2—Policy-aware matching. Can a system-derived $S = (M, T)$ policy preserve legitimate technical reconstruction variation while rejecting security-critical semantic differences?

RQ3—Acceptance-region security. How vulnerable is the induced accepted region under K_0 – K_3 attackers and finite budgets?

RQ4—Template/privacy leakage. What semantics can be inferred from stored records, policies, and authentication transcripts under named compromise views?

RQ5—Private verification. Can the frozen policy be evaluated while hiding the semantic credential and resisting practical database-only validation and cross-domain linkage under stated assumptions?

RQ6—Practicality. What computation, bandwidth, latency, storage, trust, and deployment assumptions are required?

8 Proposed methodology and baselines

The execution order is deliberately staged.

Novelty and preregistration. P8 completed a focused primary-source novelty refresh and froze graph tasks, controlled concepts, splits, capability screening, policy derivation, attack budgets, uncertainty, and numeric gate thresholds. It performed no model acquisition, dataset acquisition, v2 inference, sealed-v1 access, or cryptographic implementation.

Representation before authentication. P9-v2 screened two pre-registered monolithic extractors on 96 controlled images. Both failed the schema-validity prerequisite under exact logical futility, and reconstruction was not executed. P9-v3A is a new architecture/preregistration phase, not a repair. It freezes $I = G(P, r)$, $O = \text{Observe}(I)$, $S = C(O)$ and $(M, T) = \Pi(S)$; two modular pipelines; a new 240-image capability design; compiler/perception metrics; and Gates V3-A1/V3-A2. The pre-output v3.1 addendum changes only prospective suitability details: EGTR replaces SGTR, support is opportunity-based, and installed GPU capacity is not confused with measured pipeline consumption. Neither phase runs a model.

Policy and attackers. P10 compares dense similarity, global fuzzy overlap, a feasible VSA-style policy, and the proposed system-derived M/T policy. P11 evaluates the full acceptance region under budgeted attackers before any cryptographic implementation.

Private verification. Only a constructive semantic-security gate permits P12 to define the ideal functionality and compare mature constructions. Minimal POCs precede selection. The selected protocol is then evaluated for reference correctness, leakage, offline validation, compromise, linkage, and cost.

Practicality. RQ6 measures client-side generator, perception, compiler, and protocol costs rather than assuming them. P9-v3B has not established actual resource requirements. A powerful research GPU is experimental infrastructure, not automatically part of the intended deployment architecture. If trusted-client execution is impractical, the valid result is a deployment limitation, restricted-use outcome, failed practicality gate, or negative finding—not migration of secret-bearing prompt, image, observation, or graph processing to an untrusted cloud.

Baselines. Planned baselines include direct-text processing, dense visual/text similarity, global fuzzy semantic overlap, VSA semantic-policy behaviour, the full VSA binding model where reproducible, plaintext/hash records, and relevant mature private-matching constructions. Any inaccessible or underspecified VSA behaviour will be documented rather than guessed.

9 P9 capability outcome

The P9 capability manifest contains 64 development and 32 validation fixtures, equally divided between procedural/composite and researcher-authored synthetic-raster scenes. It accesses no participant or sealed-v1 data. Moondream2 at revision prefix 9a7d40240508 emitted no JSON on formal validation fixture cap-v2-064 in 157.57 seconds with 4.58 GiB peak process RSS. SmolVLM2-2.2B-Instruct at revision prefix c0a7af506d0f emitted malformed, truncated JSON in 392.10 seconds with 5.62 GiB peak RSS. The artifact records both complete immutable revisions.

The preregistration did not specify an early-stop rule. After separate plumbing smoke and before formal validation output, we implemented exact logical futility: because all checks are conjunctive, one invalid result makes the 98% schema-validity check impossible even if all remaining fixtures pass. This procedural deviation cannot create a pass, but it limits the negative evidence. We did not estimate full atom F1, bootstrap intervals, determinism, structured error strata, or full-set latency medians. We made no candidate replacement, prompt/output repair, fine-tuning, or authentication-outcome selection.

10 P9-v3 modular extraction preregistration

The v2 failure conflated serialization correctness with semantic correctness. A model can describe an image correctly yet truncate JSON, or emit valid JSON containing false facts. V3 therefore makes the observation object explicit: detections carry category, normalized box, component-local confidence and revision; attributes, unary actions, binary interactions, and scene hypotheses reference those detections and retain provenance, abstention, failure, latency and memory events. Confidence values are not compared across components or treated as calibrated probabilities.

The deterministic compiler validates versions, labels, boxes and references; resolves same-category duplicates under a frozen IoU/tie-break rule; assigns canonical identifiers; derives counts and spatial relations; normalizes inverse edges; and serializes sorted UTF-8 JSON. It returns a valid graph or a valid typed failure, never malformed output. At least 320 exact tests must pass with 100% oracle agreement, schema validity and byte repeatability. One compiler invariant failure blocks V3-A1 regardless of perception scores.

The final two pipelines test distinct assumptions. The first combines Grounding DINO Tiny with SigLIP 2 Base for entity localization and closed-label crop, pair-crop and scene scoring; pair-crop relation classification is an experimental compositional hypothesis, not an established relation detector. The second combines EGTR(VG) graph-native object/relation/connectivity tensors with SigLIP 2 attributes and scenes. Exact revisions, checkpoint identities, acquisition hashes, adapters, label intersections, development-only thresholds, and failure rules are preregistered. Any documented dedicated research GPU may be used, but measured complete-pipeline peak VRAM remains at most 24 GiB and peak RSS at most 32 GiB; median latency remains at most 30 seconds/image and p95 at most 60 seconds/image. Installed capacity is not a gate and cannot raise consumption limits.

Future P9-v3B uses 120 controlled geometric/composited fixtures and 120 new naturalistic outputs from the intended local generator. Each stratum has 24 semantic families split 12/12 between development and validation, five images per family. Labels and provenance are fixed before model output. The v3.1 primary gate types are entity, colour, count, binary interaction, and geometry relation; each has at least 60 positive and 60 applicable-negative validation opportunities per stratum without requiring more than two to four visible entities per image. Other types remain exploratory at 30/30 and are ‘not_gate_evaluable’, not post-hoc passes or failures of unrelated types. A type must also pass precision, recall, F1, coverage and family-level uncertainty separately in both strata.

Gate V3-A1 still requires one complete pipeline with eligible entity plus two further types, including a structural type; cross-pipeline union is forbidden. Naturalistic image creation is blocked until two independent model-blind human annotators are confirmed.

P9-v3C is future and requires a separate reconstruction freeze. Gate V3-A2 requires eligible enrolment, FRR, targeted-neighbour FAR, random FAR, same/near AUC, and a material paired M/T improvement over the strongest baseline. Passing it would establish only technical reconstruction feasibility. P10 remains blocked until V3-A2; cryptography remains further downstream.

11 Decision gates

Gate V2-N passed with narrowing. Gate V2-A is the immutable failed monolithic path. The explicit new-version path uses V3-A1 modular extraction viability and V3-A2 independent reconstruction viability before the retained V2-B policy, V2-C acceptance-region, V2-D private-functionality, V2-E protocol/privacy, and V2-F end-to-end gates. No later cost may bypass an earlier failure. Valid outcomes include standalone authentication, second factor, policy-constrained credential, measurement/negative contribution, or stop.

12 Formative v1 evidence

The completed P0–P7 programme evaluated a different system, visual-semantic-pipeline-v1. P6’s training-selected weighted matcher had validation FRR 20.8%, targeted-neighbour FAR 75.0%, and random FAR 3.0%; all tested representations failed the original gate. Twelve v1 test families remain sealed. P7 then found no uncertainty-supported material benefit from the tested SD-Turbo/Florence/SigLIP image pathways and removed that image stage from the v1 authentication core.

These are immutable formative negative results, not v2 measurements. They motivate explicit security anchors and independent capability screening but may not tune the v2 policy, representation, data, or gates. The v2 hypothesis requires new identifiers, preregistration, and evidence.

13 Expected contributions and limitations

If all gates pass, C1 would establish technical reconstructability of the scoped free-language generative path; C2 would establish only the measured incremental value of the frozen system-derived typed policy; C3 would characterise the accepted region under finite-budget attackers; and C4 would establish construction-specific private and unlinkable verification. C1/C2 are enabling contributions; C3/C4 carry the prospective principal security/privacy contribution. Their scientific meanings are unchanged by the narrative reframe. P8 rejected C5 as standalone and retained adaptive semantic verification-oracle leakage as mandatory evidence inside C3/C4.

The frozen v2 path stopped at the first risk: neither tested extractor met the structured-output prerequisite. P9-v3A provides an independently justified representation architecture but no empirical rescue. Whether either modular pipeline works remains unresolved, as do reconstruction, policy, guessing-region, private-construction, and client-side practicality questions. A research GPU does not establish deployment feasibility. The controlled methodology cannot

establish human memorability, recall, usability, preference, accessibility, natural secret-selection entropy, or real-world authentication time; nor can it establish superiority to passwords, passkeys, biometrics, or hardware authenticators. A future participant study would require separate ethics, design, and evidence.

14 Conclusion

Semantic Secrets asks what follows when authentication moves from exact reproduction to approximate semantic reconstruction. Visual generation is a testable reconstruction/normalisation boundary, not a source of entropy or an asserted memory aid. The induced acceptance region is the central security object, and privacy is a separate construction-specific property rather than a consequence of hashing semantics. P9-v2 falsified its frozen monolithic extractor path, and that evidence remains unchanged. P9-v3A supplies the explicit reframe; v3.1 prospectively freezes the final two-pipeline and support design before output. It passes no empirical gate. P9-v3B is the next permitted experiment only after explicit authorisation and resolution of the annotation blocker; P9-v3C, P10, attacker evaluation, and cryptographic prototyping remain blocked by their gates.

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